

Results Report Phase T — FusionMINERful (hybrid: PnWa + Declare)

Theoretical a-priori assessment from specification and documentation. Format follows the example reports; question wording at the level of the questionnaire update 2026-07-01.

Answers drawn from the characteristics of the fusion pipeline (imperative PNwA backbone from Heuristics/Inductive plus declarative MINERful layer) and from the tooling documentation of the chosen setup.

Structured

Item	Answer	Rationale
T-S-BQ-1 Active restriction beyond the recorded traces, rather than mere log reproduction?	Yes	The hybrid pipeline applies cut/pruning restriction.
T-S-BQ-2 Documented guarantee that the observed variants remain replayable?	No	Evaluator answer No (deviating from the seed): despite the PNwA core, no documented replay guarantee.
T-S-SF-1 Explicit, connected control-flow structure or losslessly equivalent notation?	Yes	The hybrid inherits the directed imperative backbone.

Semi-Structured

Item	Answer	Rationale
T-Sm-BQ-1 Can it enforce fixed local orderings on part of the behavior?	Yes	The PNwA backbone fixes local sequence ordering on sub-regions; the declarative layer leaves other parts open.
T-Sm-BQ-2 Can it leave part of the behavior open/unordered?	Yes	The declarative layer leaves parts of the behavior open alongside the ordered backbone.
T-Sm-SF-1 Fixed and open dependencies side by side in one model?	Yes	Hybrid with zoning component (Declare/IM fusion) represents fixed and open bindings side by side.

Loosely Structured

Item	Answer	Rationale
T-L-BQ-1 Evidence-driven individual decision per restriction (only with sufficient log support)?	Yes	The declarative layer decides each constraint individually via support thresholds.
T-L-BQ-2 Documented generalization, so that the output accepts behavior beyond the exact traces?	Yes	The hybrid generalizes via the imperative backbone plus declarative layer.
T-L-SF-1 Binding via local constraints rather than a fixed path structure?	Yes	The hybrid carries a declarative constraint layer.

Tfit Result

Structuredness class	#Yes / #Items	Tfit (%)
Structured	2/3	66.7
Semi-Structured	3/3	100.0
Loosely Structured	3/3	100.0

Interpretation. The a-priori profile is tilted toward the unstructured pole (66.7% / 100% / 100%). The hybrid combines an imperative PNwA backbone with a declarative constraint layer; on paper it thus covers, on the semi middle, both reframed capabilities — fixing local ordering via the backbone (Sm-BQ-1) and leaving other parts open via the constraint layer (Sm-BQ-2) — as well as the representation of fixed and open bindings side by side (Sm-SF-1). At the loose pole it carries the evidence-driven individual decision per constraint (L-BQ-1), the generalization (L-BQ-2), and the constraint-based notation (L-SF-1). Only the documented replay guarantee on structured logs (S-BQ-2) was negated. This a-priori strong profile is to be tested against the empirical output (Phase E).

Note on the setup. The empirical runs (Phase E) use the Phase B setup (`activity_entropy 0.5`, `dependency_threshold 0.9`, `declare_support 1.0`, `im_fitness 0.2`, `resilience 0.1`). The a-priori assessment refers to the documented pipeline capability.

Results Report — Phase E (empirical), FusionMINERful (hybrid) · all three structuredness classes · BQ & IN

1 Objective and procedure

The models discovered with FusionMINERful are evaluated against the empirical Phase E items of all three classes. The miner delivers a hybrid model: a Petri net with annotations (PnWa) as the imperative backbone, augmented by a declarative constraint layer. Four items per class: two on Behavioral Quality (BQ), two on Interpretability (IN). Three-point scale per item (2 = full, 1 = partial, 0 = absent). Each item is averaged over the three logs of the class, then the class Efit is formed: $Efit = \text{sum of item means} / 8 \times 100$.

Note on the metrics. Fitness is the PNwA replay fitness (mpcc), which — unlike the purely imperative miners — is not guaranteed to be 1.0 (Log04: 0.444). Precision is alignment-based on the plain net. The *Constr.* column counts the declarative constraints of the hybrid layer: where it is 0, the hybrid collapses to a purely procedural PnWa net.

2 Data basis: metrics of all models

Log	Cl.	Fitness	Precision	Gen.	Constr.	PnWa-Trans	Edges	CFC
Log01_structured	S	0.781	0.946	0.999	8	7	10	2
Log04_structured	S	0.444	1.000	1.000	16	6	6	2
Log05_structured	S	1.000	0.953	0.999	0	6	12	6
Log02_semiStructured	Sm	0.830	0.570	0.999	10	9	22	5
Log06_semiStructured	Sm	0.760	0.709	0.999	28	10	24	3
Log07_semiStructured	Sm	0.900	0.545	0.998	22	11	30	5
Log03_looselyStructured	L	1.000	0.744	0.999	0	14	34	12
Log08_looselyStructured	L	0.848	0.783	0.999	5	13	26	9
Log11_looselyStructured	L	1.000	0.543	1.000	0	8	16	8

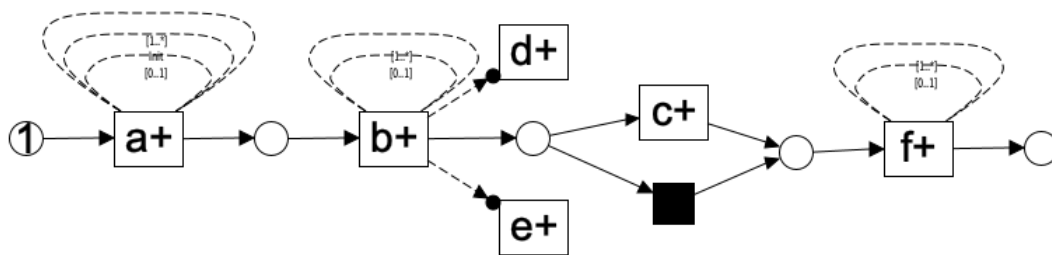
Fitness = PNwA replay fitness (mpcc). Gen. = generalization. Constr. = number of declarative constraints in the hybrid layer. CFC = control-flow complexity. The Log11 run was reproduced with the Phase B setup (model image rendered by hand from the PnWa net, since the ProM renderer produces no image headless).

3 Class Structured — Efit = 62.5%

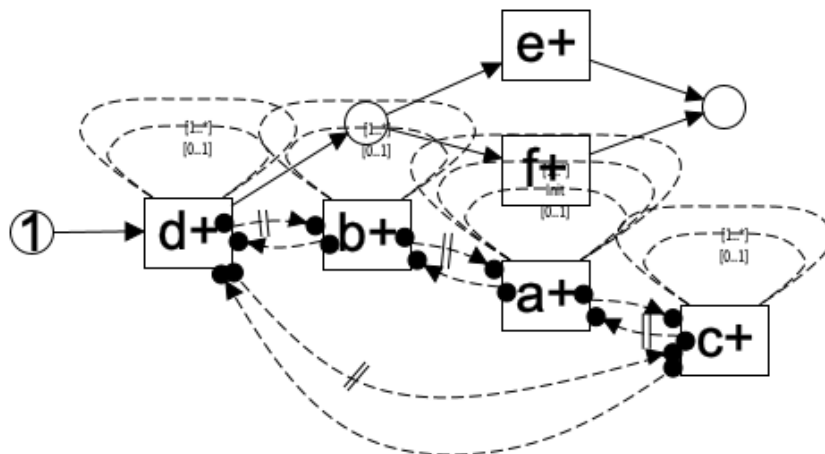
Logs: Log01, Log04, Log05. Behavioral Quality 75.0% · Interpretability 50.0%.

Item	Challenge	Log01	Log04	Log05	Mean	Question (short)
E-S-BQ-1	Behav. Quality	1	1	2	1.33	How tightly does the output bound the behavior space beyond the observed variants?
E-S-BQ-2	Behav. Quality	2	1	2	1.67	Do the variants attested in the log remain replayable?
E-S-IN-1	Interpret.	1	0	2	1.00	How compactly does the output represent the behavior?
E-S-IN-2	Interpret.	1	0	2	1.00	Can the execution ordering be read off from the output?
Efit sum					5.00/8	= 62.5%

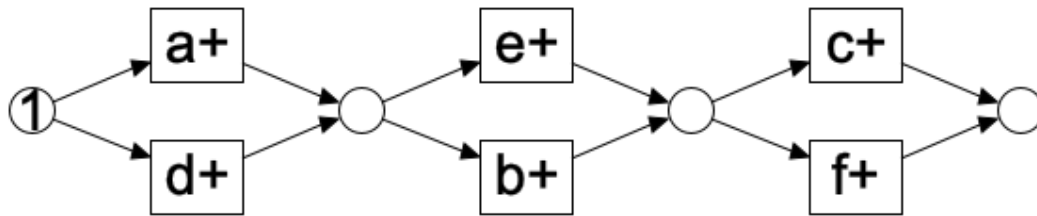
Rationale per item. E-S-BQ-1: despite tight precision (0.95 / 1.00 / 0.95) the declarative constraint layer leaves more behavior open than the net image suggests — for Log01, d may follow f; for Log04, the run may also start with d. Log01 and Log04 therefore at 1; Log05 (no constraints, compact) at 2. E-S-BQ-2: Log01 and Log05 are fully replayable (fitness 0.78 resp. 1.00 with tight precision) → 2. Log04 falls to 1: 16 constraints push the PNwA fitness to 0.444, attested variants no longer run through. E-S-IN-1: Log05 is compact without constraints (CFC 6) → 2. Log01 carries 8 constraints → 1. Log04 is overloaded with 16 constraints on six transitions → 0. E-S-IN-2: Log05 shows a clear left-to-right flow → 2. Log01 is only partly readable as an ordered model due to the constraint overlay → 1. Log04, with 16 constraints on six transitions → 0.



Log01 · Structured · PnWa net



Log04 · Structured · PnWa net



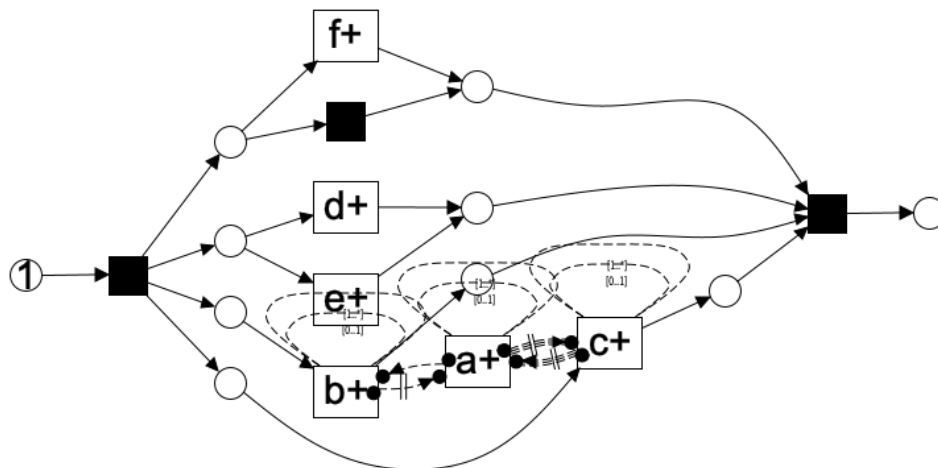
Log05 · Structured · PnWa net

4 Class Semi-Structured — Efit = 16.7%

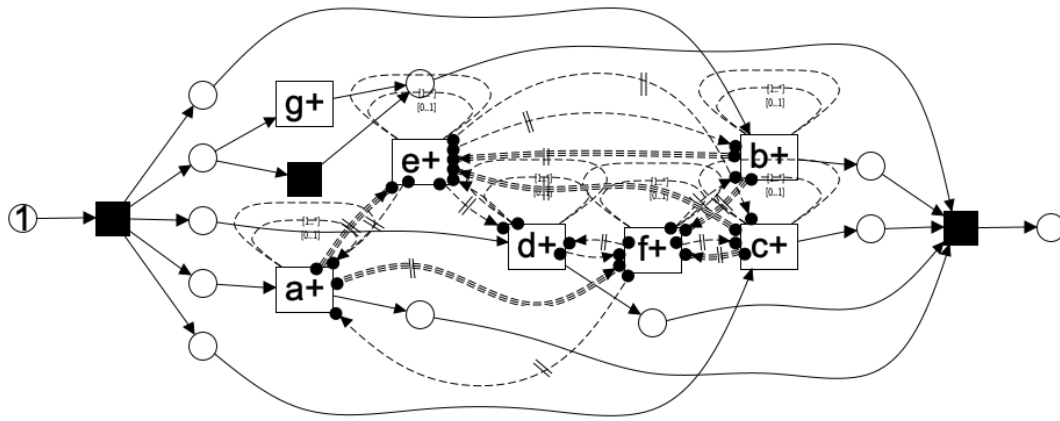
Logs: Log02, Log06, Log07. Behavioral Quality 33.3% · Interpretability 0.0%.

Item	Challenge	Log02	Log06	Log07	Mean	Question (short)
E-Sm-BQ-1	Behav. Quality	1	0	1	0.67	Does the model fix the internal ordering of the stable fragments?
E-Sm-BQ-2	Behav. Quality	1	0	1	0.67	Does the model allow fragment recombinations beyond the observed arrangement, with replay preserved?
E-Sm-IN-1	Interpret.	0	0	0	0.00	Are the stable and the flexible reference regions distinguishable?
E-Sm-IN-2	Interpret.	0	0	0	0.00	Is the output readable as one coherent model spanning both regions?
Efit sum					1.33/8	= 16.7%

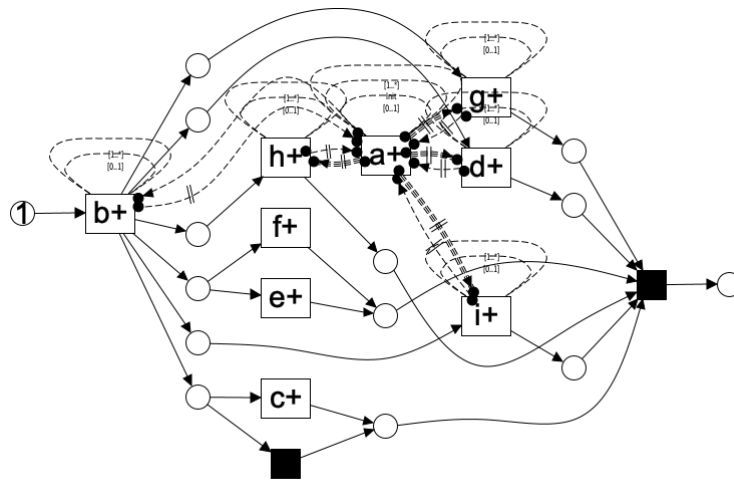
Rationale per item. E-Sm-BQ-1: Log02 and Log07 recognize individual fragments (d/e/f/g partly visible) and order them partly firmly, but apply the restriction at the wrong places via Petri net and constraints → 1 each. Log06 is almost uniform, every activity carries constraints, no fixed fragment ordering recognizable → 0. E-Sm-BQ-2: Log02/Log07 replay attested variants for the most part (fitness 0.83 / 0.90) and admit some recombination → 1 each. Log06 cannot be meaningfully judged; the fitness value alone carries no statement → 0. E-Sm-IN-1: all 0. The stable and flexible regions are not separable: without knowledge of the ARM fragments the relations cannot be read off, the PnWa is overloaded with 22–28 constraints and crossing lines. E-Sm-IN-2: all 0. A large AND-split plus many constraints make the output not readable as one coherent model spanning both regions.



Log02 · Semi-Structured · PnWa net



Log06 · Semi-Structured · PnWa net



Log07 · Semi-Structured · PnWa net

5 Class Loosely Structured — Efit = 33.3%

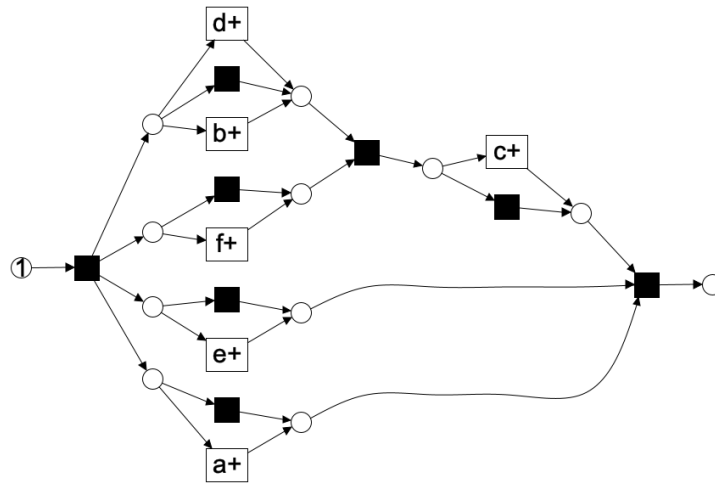
Logs: Log03, Log08, Log11. Behavioral Quality 25.0% · Interpretability 41.7%.

Note on the assessment. Log11 was run with the Phase B setup; the PnWa net (below) confirms the metrics (fitness 1.0, precision 0.543, generalization 1.0, 0 constraints, 8 transitions, 16 edges, CFC 8) and the flower-hub character around a central place. Log08 IN-2 was supplemented from the model. The remaining loosely values come from the stored assessment.

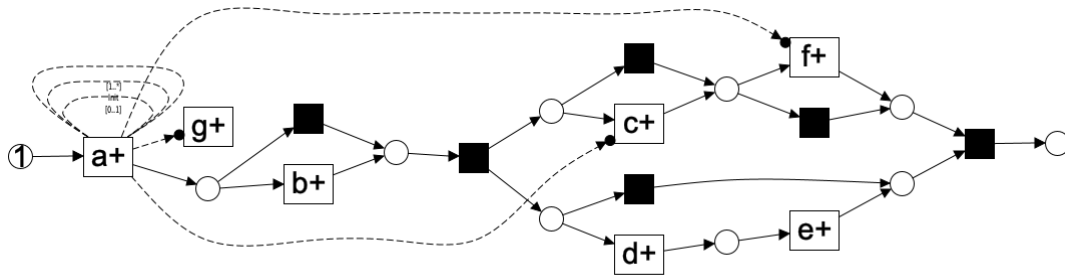
Item	Challenge	Log03	Log08	Log11	Mean	Question (short)
E-L-BQ-1	Behav. Quality	0	1	0	0.33	Does the model impose local restrictions that clearly exclude unsupported behavior?
E-L-BQ-2	Behav. Quality	0	2	0	0.67	Does the model allow behavior beyond the exactly observed variants, with replay preserved?
E-L-IN-1	Interpret.	0	1	1	0.67	Does the output condense to a few essential regularities?
E-L-IN-2	Interpret.	1	1	1	1.00	Does the output make local dependencies readable without forcing a global ordering?
Efit sum					2.67/8	= 33.3%

Rationale per item. E-L-BQ-1: Log03 is fully permissive, practically every activity optional (precision 0.744, but everything skippable) → 0. Log11 (precision 0.543, flower hub around a central place, 0 constraints) restricts even less → 0. Log08 shows individual fixed activities (a, d/e), yet much stays optional and skippable → 1. E-L-BQ-2: Log08 carries five declarative constraints that bound plausible behavior beyond the exact variants → 2. Log03 and Log11 have 0 constraints;

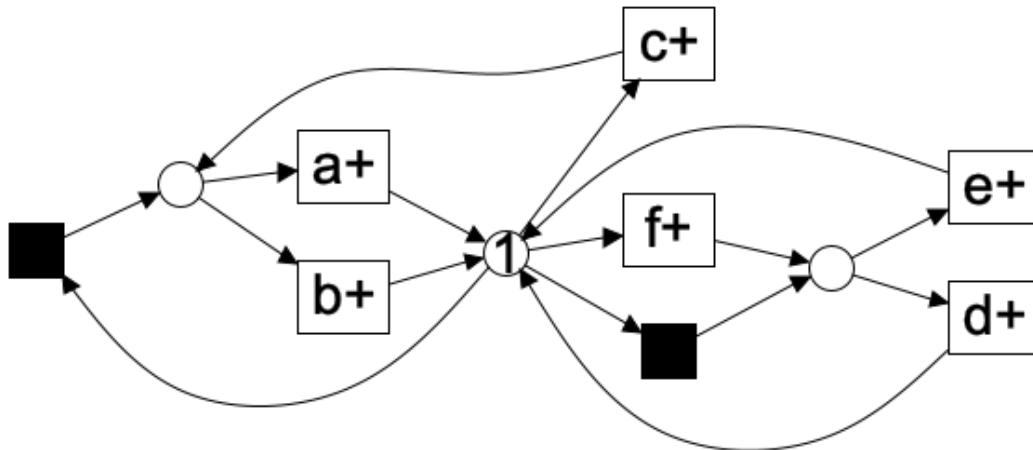
their permissiveness is vacuous, no directed generalization $\rightarrow 0$ each. E-L-IN-1: Log08 (CFC 9) and Log11 (CFC 8, most compact net) condense to a manageable core $\rightarrow 1$ each. Log03 takes in all activities without exception (CFC 12, 34 edges) $\rightarrow 0$. E-L-IN-2: all three force no global ordering (large AND-split resp. hub) and let local dependencies be partly read off $\rightarrow 1$ each.



Log03 · Loosely Structured · PnWa net



Log08 · Loosely Structured · PnWa net



Log11 · Loosely Structured · PnWa net

6 Comparison of the classes

Class	BQ	IN	Efit
Structured	75.0%	50.0%	62.5%
Semi-Structured	33.3%	0.0%	16.7%
Loosely Structured	25.0%	41.7%	33.3%

BQ = mean of the two behavioral-quality items, normalized to the scale ceiling. IN = mean of the two interpretability items, normalized to the scale ceiling.

7 Classification

The empirical partial score is not monotonic in structuredness: 62.5% → 16.7% → 33.3%. The low point is in the semi middle, not at the loose pole.

Structured (62.5%): on the behavioral side (75.0%) precision is tight (0.95–1.00), yet the declarative layer leaves more behavior open than the net image suggests (Log01: d possible after f; Log04: start also with only d). Interpretability (50.0%) falls further back because the constraint layer already produces ballast here: Log04 carries 16 constraints on six transitions, pushes fitness to 0.444, and no longer permits any ordering to be read off.

Semi-Structured (16.7%): the actual weakness. Exactly the class that the hybrid's a-priori profile addresses collapses empirically. The PnWa is overloaded with 22–28 constraints and crossing lines; stable and flexible regions are not separable (IN uniformly 0), and the restriction sits at the wrong places. The promised advantage of a differential, zone-bound filtering does not translate into the model image.

Loosely Structured (33.3%): mixed. On the behavioral side (25.0%) the hybrid is weak: without declarative constraints (Log03, Log11) the output becomes a permissive, vacuous core; only where constraints are preserved (Log08, five of them) does a bound generalization emerge. Interpretability (41.7%) is higher than in the semi middle because the nets are smaller and force no global ordering. Log11 exemplifies the collapse into a flower hub (precision 0.543).